

On Biadjoint Triangles

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May 29, 2026

CONTENTS

1	Monads	1
1.1	Monads and their Algebras	1
1.2	The Adjoint Triangle Theorem	5

1 MONADS

1.1 Monads and their Algebras

It would be shameful not to begin our exploration without a good background in monads, which are described in detail in this section. Monads, loosely speaking, answer the question of what it means for a category to possess some *algebraic* structure. This quality is made more explicit in the definition of *monadicity*, which cements the idea that a category is *monadic* over another category if it has some algebraic quality in relation to the latter category. This intuition becomes all the more clear once examples of monads and monadicity are studied. Much of this comes directly from Mac Lane, and will begin with a quick remark on composeability of functors and natural transformations.

For a category $\mathbb{X}$ and endofunctor $T : \mathbb{X} \rightarrow \mathbb{X}$, there are composite endofunctors $T^2 = T \circ T$ and $T^3 = T^2 \circ T$. If, additionally, there is a natural transformation $\mu : T^2 \rightarrow T$, then, for each object x in $\mathbb{X}$, the natural transformations $T\mu, \mu T : T^3 \rightarrow T^2$ have, respectively, components $(T\mu)_x = T\mu_x$, and $(\mu T)_x = \mu_{Tx}$.

Definition 1.1. [2] A **monad** $T = (T, \eta, \mu)$ on a category $\mathbb{X}$ is a triple in which $T : \mathbb{X} \rightarrow \mathbb{X}$ is an endofunctor on $\mathbb{X}$ and η, μ are two natural transformations

$$1_{\mathbb{X}} \xrightarrow{\eta} T \xleftarrow{\mu} T^2 \quad (1.1)$$

such that the following diagram commutes:

$$\begin{array}{ccccc} T^3 & \xrightarrow{\mu T} & T^2 & \xleftarrow{\eta T} & T \\ T\mu \downarrow & & \mu \downarrow & \swarrow & \downarrow T\eta \\ T^2 & \xrightarrow{\mu} & T & \xleftarrow{\mu} & T^2 \end{array} \quad (1.2)$$

Replace in Theorem 1.1 the endofunctor $T : \mathbb{X} \rightarrow \mathbb{X}$ with a set M , T^2 and T^3 with the cartesian product $M \times M$ and $M \times M \times M$ respectively, and lastly the natural transformations $\eta : 1_{\mathbb{X}} \rightarrow T$ with $\eta : 1 \rightarrow M$ and $\mu : T^2 \rightarrow T$ with the binary operation $\mu : M \times M \rightarrow M$. Then the diagram in Equation (1.2) expresses the usual axioms of a monoid. This is what motivates the naming of the natural transformations η as the **identity** and μ as the **multiplication** of a monad (T, η, μ) . This is also where we recover the famous “a monad is just a monoid in ...”

Definition 1.2. Let $T = (T, \eta, \mu)$ be a monad on $\mathbb{X}$. A **T -algebra** is a pair (x, ξ) in which x is an object in $\mathbb{X}$ and $\xi : Tx \rightarrow x$ is a morphism in $\mathbb{X}$ making the diagram

$$\begin{array}{ccccc} T^2x & \xrightarrow{\mu_x} & Tx & \xleftarrow{\eta_x} & x \\ T\xi \downarrow & & \xi \downarrow & \swarrow & \\ Tx & \xrightarrow{\xi} & x & & \end{array} \quad (1.3)$$

commute. A **morphism of T -algebras** $h : (x, \xi) \rightarrow (x', \xi')$ is a morphism $h : x \rightarrow x'$ in $\mathbb{X}$ making the diagram below commute:

$$\begin{array}{ccc} Tx & \xrightarrow{Th} & Tx' \\ \xi \downarrow & & \downarrow \xi' \\ x & \xrightarrow{h} & x' \end{array} \quad (1.4)$$

Definition 1.3. For a monad $T = (T, \eta, \mu)$ on $\mathbb{X}$, the **Eilenberg-Moore category of T -algebras**, denoted by **$T\text{-Alg}$** , is the category of T -algebras with T -algebra morphisms.

Theorem 1.4. Every adjunction $(F \dashv U, \eta, \epsilon) : \mathbb{X} \rightarrow \mathbb{A}$ determines a monad $(UF, \eta, U\epsilon F)$ on $\mathbb{X}$.

Proof. Firstly, $T = UF$ is an endofunctor on $\mathbb{X}$, η is a natural transformation $\eta : 1_{\mathbb{X}} \rightarrow UF = T$ and $\mu = U\epsilon F : UFUF = T^2 \rightarrow T = UF$ is a natural transformation. For an object x in $\mathbb{X}$, the commutativity of the diagram below must be checked:

$$\begin{array}{ccccc}
 UFUFUFx & \xrightarrow{U\epsilon_{FUFx}} & UFUFx & \xleftarrow{\eta_{UFx}} & UFx \\
 \downarrow UFU\epsilon_{Fx} & & \downarrow U\epsilon_{Fx} & \swarrow & \downarrow UF\eta_x \\
 UFUFx & \xrightarrow{U\epsilon_{Fx}} & UFx & \xleftarrow{U\epsilon_{Fx}} & UFUFx
 \end{array} \tag{1.5}$$

The left square commutes because of the naturality of $\epsilon : FU \rightarrow 1_{\mathbb{A}}$ applied to $\epsilon_{Fx} : FUFx \rightarrow Fx$, which is preserved under U . The right square commutes because of the triangle identities for the adjunction $(F \dashv U, \eta, \epsilon) : \mathbb{X} \rightarrow \mathbb{A}$. The identities are recalled:

$$\begin{array}{ccc}
 F & \xrightarrow{F\eta} & FUF \\
 & \searrow & \downarrow \epsilon F \\
 & & F
 \end{array}
 \quad
 \begin{array}{ccc}
 UFU & \xleftarrow{\eta U} & U \\
 \downarrow U\epsilon & & \swarrow \\
 U & &
 \end{array} \tag{1.6}$$

This is shown by first considering the bottom triangle in Equation (1.5), which is just the x -component of the left triangle of Equation (1.6) under U , and so it commutes. The top triangle in Equation (1.5) is the Fx -component of the right triangle in Equation (1.5). This shows that $(UF, \eta, U\epsilon F)$ is a monad on $\mathbb{X}$. This monad is called the **monad defined by the adjunction** $(F \dashv U, \eta, \epsilon) : \mathbb{X} \rightarrow \mathbb{A}$. $\square$

Then, naturally, the question arises of whether every monad induces an adjunction. The answer is *yes*, and is discussed in the next theorem.

Theorem 1.5. Every monad $T = (T, \eta, \mu)$ on $\mathbb{X}$ determines an adjunction $(F^T \dashv U^T, \eta^T, \epsilon^T) : \mathbb{X} \rightarrow T\text{-Alg}$. The monad defined by this adjunction is the same monad $T = (T, \eta, \epsilon)$.

Proof. Define

1. the right adjoint $U^T : T\text{-Alg} \rightarrow \mathbb{X}$ to be the forgetful functor, $U^T(x, \xi) := x$;
2. the left adjoint $F^T : \mathbb{X} \rightarrow T\text{-Alg}$ to be the free functor, $F^T x := (Tx, \mu_x)$;
3. the unit $\eta^T : 1_{\mathbb{X}} \rightarrow U^T F^T$ to be the same η of the monad;
4. the counit $\epsilon^T : F^T U^T \rightarrow 1_{T\text{-Alg}}$ to be $\epsilon^T_{(Tx, \mu_x)} := \mu_x$.

Now it is checked that this is indeed the desired adjunction. Firstly, note that (Tx, μ_x) is a T -algebra. That is, the diagram from Equation (1.3) commutes:

$$\begin{array}{ccc}
 T^3 x & \xrightarrow{\mu_{Tx}} & T^2 x \xleftarrow{\eta_{Tx}} Tx \\
 \downarrow T\mu_x & & \downarrow \mu_x \\
 T^2 x & \xrightarrow{\mu_x} & Tx
 \end{array}$$

But this is just the x -component of Equation (1.2), which commutes because T is a monad.

Now to check naturality of η^T and ϵ^T . The naturality of η^T is just the naturality of η . This is shown by taking x an object in $\mathbb{X}$, then the x -component of η^T is the morphism in $\mathbb{X}$, $\eta_x^T : x \rightarrow U^T F^T x = U^T(Tx, \mu_x) = Tx$, which is the same as η_x . Now for ϵ^T : take (x, ξ) an object in $T\text{-Alg}$. Then $F^T U^T(x, \xi) = F^T x = (Tx, \mu_x)$. Then the (x, ξ) -component of ϵ^T is $\epsilon_{(x, \xi)}^T : (Tx, \mu_x) \rightarrow (x, \xi)$. Taking a morphism of $h : (x, \xi) \rightarrow (x', \xi')$ in $T\text{-Alg}$ gives a naturality square:

$$\begin{array}{ccccc} (x, \xi) & & (Tx, \mu_x) & \xrightarrow{\mu_x} & (x, \xi) \\ h \downarrow & & F^T U^T h \downarrow & & \downarrow h \\ (x', \xi') & & (Tx', \mu_{x'}) & \xrightarrow{\mu'_{x'}} & (x', \xi') \end{array} \quad (1.7)$$

For some morphism $f : x \rightarrow x'$ in $\mathbb{X}$, $F^T f : (Tx, \mu_x) \rightarrow (Tx', \mu_{x'})$ must be a morphism of T -algebras, which means $F^T f : Tx \rightarrow Tx'$ must be defined as $F^T f := Tf$. Then $F^T U^T h = Th$. So $F^T U^T h$ in Equation (1.7) becomes Th , and the square thus commutes by naturality of μ , and because h is a morphism of T -algebras. To check that there is an adjunction is to check the triangles in Equation (1.6) commute. For (x, ξ) an object in $T\text{-Alg}$ and x an object in $\mathbb{X}$, the left triangle is the (x, ξ) -component of the left triangle of Equation (1.7), and the right is the x -component of the right triangle of Equation (1.7):

$$\begin{array}{ccc} Tx & \xrightarrow{T\eta_x} & T^2 x \\ & \searrow & \downarrow \mu_x \\ & & Tx \end{array} \quad \begin{array}{ccc} Tx & \xleftarrow{\eta_x} & x \\ \xi \downarrow & & \nearrow \\ x & & \end{array}$$

The left and right triangles commute respectively from the commutativity of the monad diagram Equation (1.2) and the T -algebra diagram Equation (1.3).

Lastly, notice that for an object x in $\mathbb{X}$, $U^T F^T x = Tx$, $\eta^T = \eta$ and $U^T \epsilon^T F^T x = \mu_x$, and so the monad defined by this adjunction is $T = (T, \eta, \mu)$, as desired. $\square$

So we have that a monad determines an adjunction, which, in return determines the same monad again. The reverse is not always true: an adjunction determines a monad which does not always determine the same adjunction. When an adjunction induces a monad that induces the same adjunction back, we call that adjunction *monadic*. We get a sense of how monadic an adjunction is (or measure its *monadicity* with the following construction.

Theorem 1.6. *Let $(F \dashv U, \eta, \epsilon) : \mathbb{X} \rightarrow \mathbb{A}$ be an adjunction and $T = (UF, \eta, U\epsilon F)$ be the monad defined by the adjunction. Then there is a unique functor $K : \mathbb{A} \rightarrow T\text{-Alg}$ with $U^T K = U$ and $KF = F^T$.*

Proof. Define $K : \mathbb{A} \rightarrow T\text{-Alg}$ by $Ka := (Ua, U\epsilon_a)$, and on morphisms $Kf := Uf$. $\square$

There is another arrow that will make the triangle complete: the functor K as defined above has a left adjoint, the functor $L : T\text{-Alg} \rightarrow \mathbb{A}$.

Theorem 1.7. *There is an adjunction $(L \dashv K, \bar{\eta}, \bar{\epsilon}) : T\text{-Alg} \rightarrow \mathbb{A}$.*

Proof. We define

1. the left adjoint $L : T\text{-Alg} \rightarrow \mathbb{A}$, where $L(x, \xi)$ is defined as the coequaliser of

$$FUFx \xrightarrow[F\xi]{\epsilon Fx} Fx \xrightarrow{\pi(x, \xi)} L(x, \xi)$$

2. the right adjoint $K : \mathbb{A} \rightarrow T\text{-Alg}$, defined by $Ka := (Ua, U\epsilon_a)$ on objects and $Kf := Uf$ on morphisms;
3. the unit $\bar{\eta} : 1_{T\text{-Alg}} \rightarrow KL$ defined by $\bar{\eta}_{(x, \xi)} = U\pi_{(x, \xi)}\eta_x$;
4. the counit $\bar{\epsilon} : LK \rightarrow 1_{\mathbb{A}}$ is defined for each $\bar{\epsilon}_a$ as the unique morphism making the diagram

$$\begin{array}{ccc} FUFUa & \xrightarrow[FU\epsilon_a]{\epsilon FUa} & FUA \xrightarrow{\pi(Ua, U\epsilon_a)} (LUa, U\epsilon_a) \\ & & \searrow \epsilon_a \quad \downarrow \bar{\epsilon}_a \\ & & a \end{array}$$

commute.

□

Now our picture is complete, and looks like below, with the F - and U -triangles commuting.

$$\begin{array}{ccc} & & \mathbb{X} \\ & \nearrow U & \\ \mathbb{A} & & \\ & \nwarrow F & \\ & & T\text{-Alg} \\ & \nwarrow F^T & \\ & & \\ & \nearrow K & \\ & \nwarrow L & \end{array}$$

Definition 1.8. Let $(F \dashv U, \eta, \epsilon) : \mathbb{X} \rightarrow \mathbb{A}$ be an adjunction and $T = (T, \eta, \mu)$ be its associated monad. Then

1. the functor $K : \mathbb{A} \rightarrow T\text{-Alg}$ as defined in the previous result is called the *comparison functor*;
2. the functor $U : \mathbb{A} \rightarrow \mathbb{X}$ is said to be *monadic* when the comparison functor is an equivalence.

Monadic functors are very useful, because they recall the adjunction from which they came. Moreover, monadicity means that, in some sense, a category is algebraic because it is equivalent to a category of algebras.

As with any free-forgetful adjunction to a category of algebras, it is useful to know what the free algebras of the category are. In the case of the adjunction

Definition 1.9.

Theorem 1.10. *there is an adhybc*

1.2 The Adjoint Triangle Theorem

Definition 1.11. [1] Given a functor $\mathbb{A} \xrightarrow{J} \mathbb{B}$, an **adjoint triangle for J** consists of a category $\mathbb{C}$, and adjunctions $(E \dashv R, \rho, \mu) : \mathbb{A} \rightarrow \mathbb{C}$ and $(L \dashv U, \eta, \varepsilon) : \mathbb{B} \rightarrow \mathbb{C}$ such that $JR = U$, displayed in the diagram:

$$\begin{array}{ccc} \mathbb{A} & \xrightarrow{J} & \mathbb{B} \\ \swarrow R & \nearrow U & \\ \mathbb{C} & \xleftarrow{L} & \mathbb{C} \\ \nwarrow E & \nearrow L & \end{array}$$

. The equality $JR = U$ is sometimes written as the natural transformation $JR \xrightarrow{1} U$ and *vice versa*.

Lemma 1.12. [1] In the notation above, given an adjoint triangle for a functor J , there is a natural transformation $LJ \xrightarrow{\Theta} E$ which makes the following diagram commute:

$$\begin{array}{ccc} JRLJ & \xrightarrow{JR\Theta} & JRE \\ \uparrow 1_{LJ} & & \uparrow J\rho \\ ULJ & \xleftarrow{\eta J} & J \end{array}$$

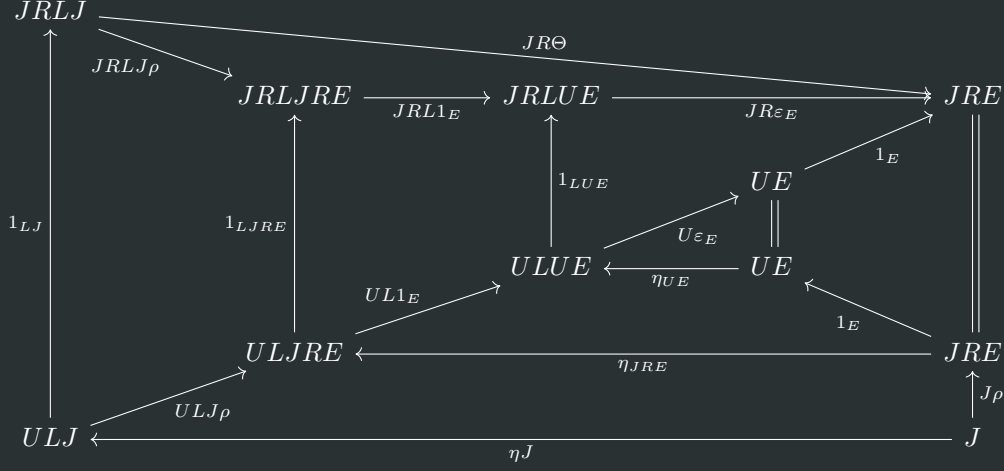
Proof. Define Θ as the composite:

$$\begin{array}{ccc} LJRE & \xrightarrow{L1_E} & LUE \\ \uparrow LJ\rho & & \downarrow \varepsilon_E \\ LJ & \xrightarrow{\Theta} & E \end{array}$$

Then we clarify the diagram:

□

$$\begin{array}{ccccccc} JRLJ & \xrightarrow{JRLJ\rho} & JRLJRE & \xrightarrow{JRL1_E} & JRLUE & \xrightarrow{JR\varepsilon_E} & JRE \\ \uparrow 1_{LJ} & & \uparrow 1_{LJRE} & & \uparrow 1_{LUE} & & \uparrow 1_E \\ & & & & & & UE \\ & & & & & & \parallel \\ & & & & & & UE \\ & & & & & & \downarrow 1_E \\ & & & & & & JRE \\ & & & & & & \uparrow J\rho \\ & & & & & & J \\ & & & & & & \uparrow \eta J \\ ULJ & \xleftarrow{ULJ\rho} & ULJRE & \xleftarrow{UL1_E} & ULUE & \xleftarrow{\eta_{UE}} & UE \\ & & & & & & \downarrow \varepsilon_E \\ & & & & & & E \end{array}$$



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